Your LLM no longer reads every token

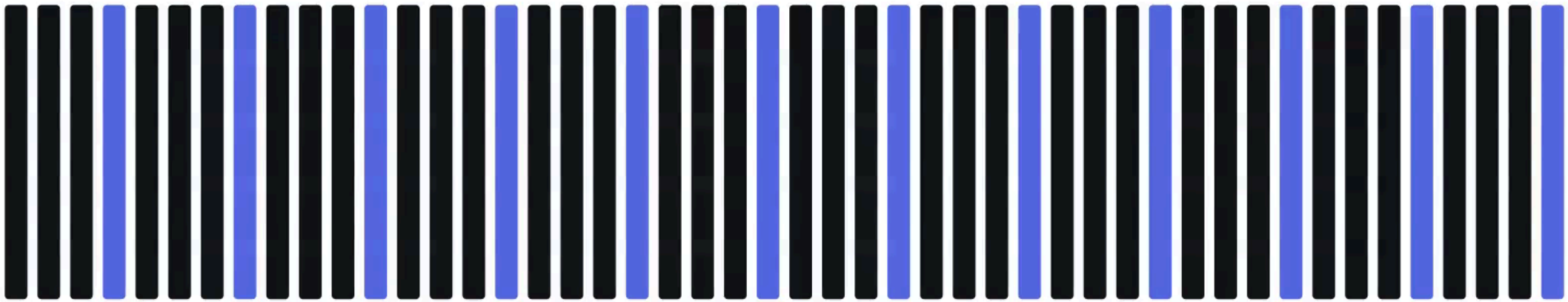
Eight attention designs, explained and measured

Julien Simon · October 2026 · companion to "Eight Changes to Transformer Attention, and What Each Costs" on airealist.ai
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AUGUST 2026 · QWEN3.8-FLASH-NEXT · config.json

Zero layers read every earlier token

48 LAYERS



0

LAYERS READING EVERY
EARLIER TOKEN IN FULL

36 × linear_attention

a fixed-size state, no per-token cache

12 × "full_attention"

indexer_budget: 2048 → reads 2,048 picked tokens

Source: huggingface.co/Qwen/Qwen3.8-Flash-Next/raw/main/config.json, read October 1st and 3rd, 2026

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Why: two bills grow with the context

- **Memory**: a full-attention layer keeps a key and a value for every token, for every user
- **Compute**: work per token grows with the context; reading a whole prompt grows with its square
- Agents resend their whole history at every step
- DeepSeek's estimate for V4-Pro at a million tokens: **10%** of V3.2's cache, **27%** of its compute per token

The KV cache from scratch: "Decoder-only inference" (2025) · MQA, GQA, windows: "Better Attention layers" (2024)

DeepSeek-V4 report, arXiv 2606.19348 · youtube.com/watch?v=cl3MAhAr9-M · youtube.com/watch?v=2TT384U4vQg

How to read every diagram in this video

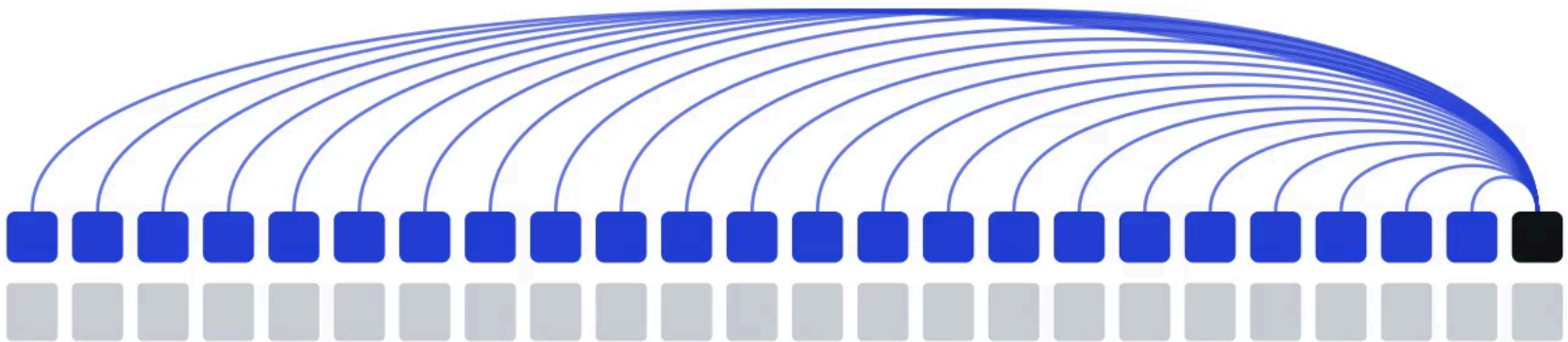
- A row of squares: the **earlier tokens**, oldest on the left
- The black square on the right: the token being **written now**
- A **blue line**: an earlier token it reads directly; blue squares are the ones it reads
- A **black box**: a fixed-size state, the same size whatever the length
- Grey cells under the tokens: the **cache** kept for each one
- On the right: the cache this design holds as tokens arrive, against full attention in grey

Hybrids get one panel per layer type, plus a strip of the model's layer mix

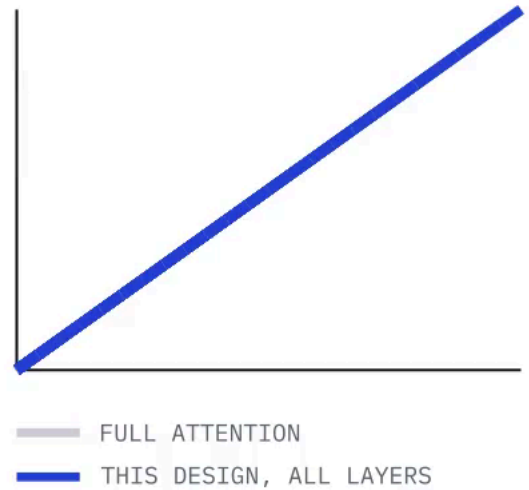
2017 · THE BASELINE

Full attention: every token reads every earlier token

EVERY LAYER



CACHE HELD AS TOKENS ARRIVE



ARCS	the newest token reads every earlier token
GREY CELLS	the cache: one key and one value kept per token
STILL SHIPS	Mistral Medium 3.5, IBM Granite 4.2, K2-Horizon

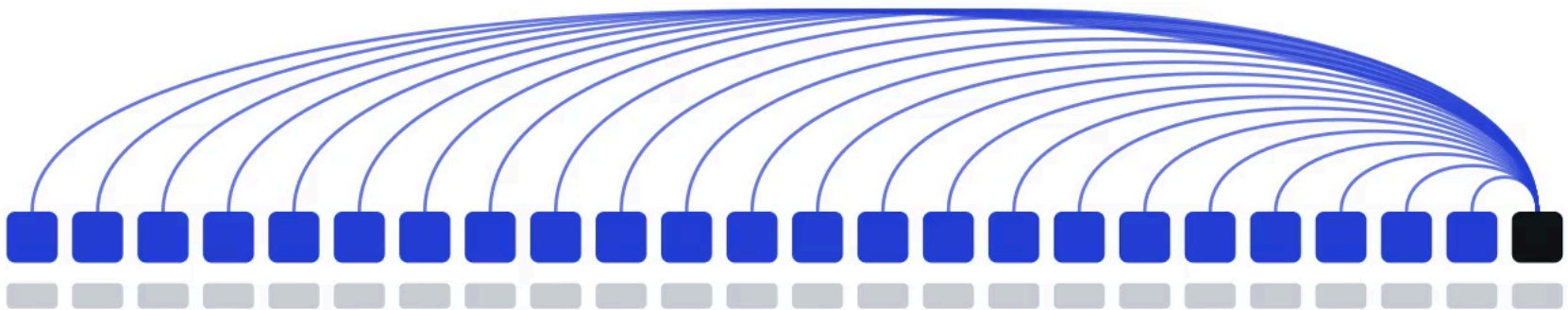
Source: Vaswani et al., 2017 · configs read October 1st, 2026

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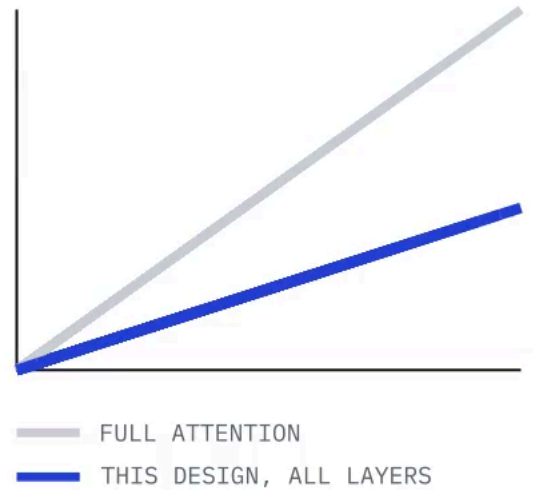
STEP 1 · 2023

Grouped-query attention: same reads, smaller cache

EVERY LAYER · HEADS SHARE KEYS AND VALUES IN GROUPS



CACHE HELD AS TOKENS ARRIVE



SAVES	memory: the cache per token shrinks by the group factor
GIVES UP	some accuracy: 41.2 against 45.2 in DeepSeek's 7B test
SHIPS IN	K2-Horizon and most dense models

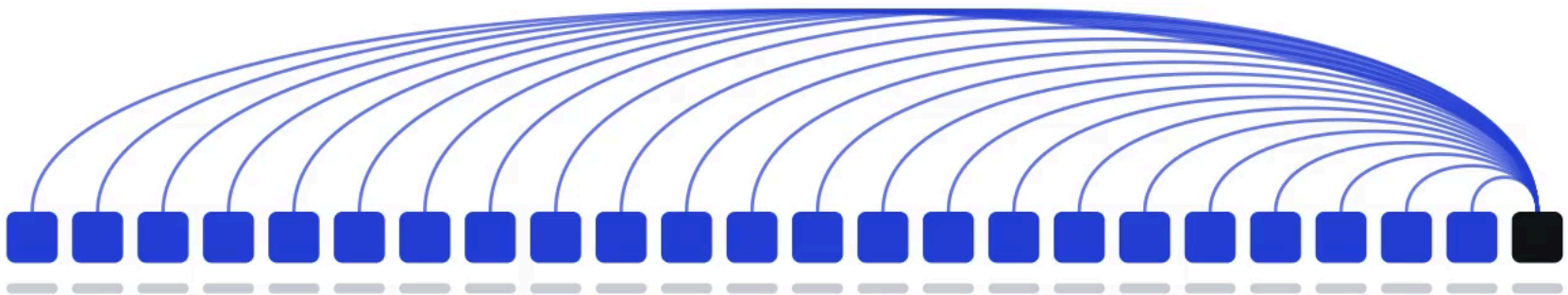
Source: Ainslie et al., 2023 · DeepSeek-V2 report, Table 8

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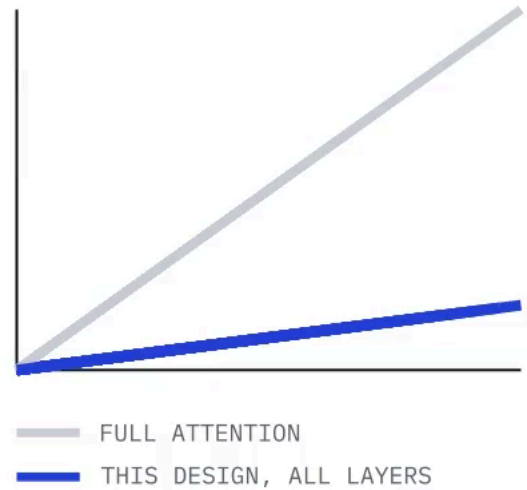
STEP 2 · MAY 2024

Latent attention: same reads, compressed cache

EVERY LAYER · ONE 576-NUMBER VECTOR PER TOKEN



CACHE HELD AS TOKENS ARRIVE



SAVES	memory: the cache at 4 to 14% of the original
GIVES UP	very little: won 7 of 8 comparisons in DeepSeek's test
SHIPS IN	DeepSeek V2 and V3, Kimi's full-attention layers

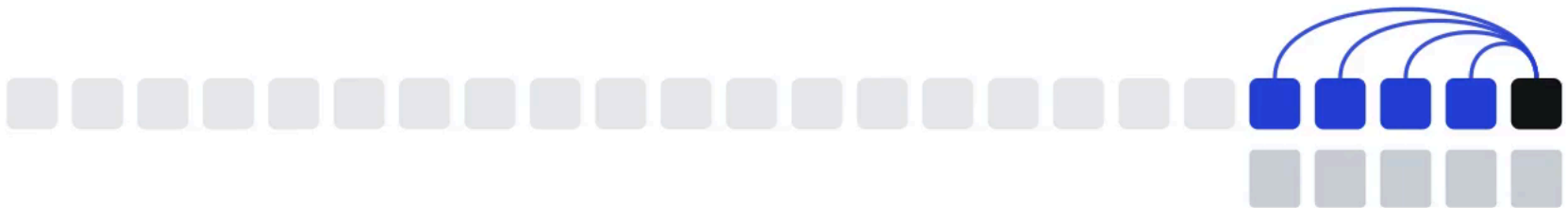
Source: DeepSeek-V2 report, May 2024

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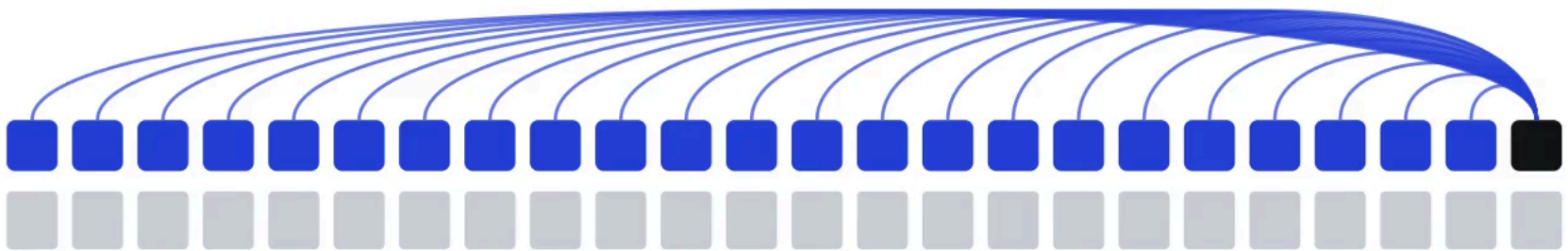
STEP 3

Sliding window: most layers read only the recent tokens

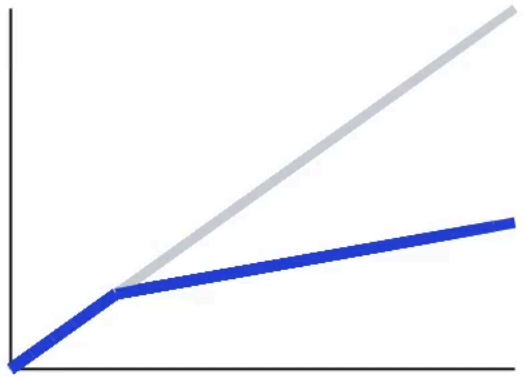
WINDOW LAYERS · THE LAST 128 TO 2,048 TOKENS



FULL LAYERS · EVERYTHING



CACHE HELD AS TOKENS ARRIVE



LAYER MIX



Muse Glimmer 3 : 1 (shown)
Gemma 4, Inkling 5 : 1
gpt-oss 1 : 1

SAVES	memory and compute in the window layers
GIVES UP	long range there: 90.0 → 72.0 at 128K in MiniMax's test
SHIPS IN	gpt-oss, Gemma 4, Inkling, Muse Glimmer

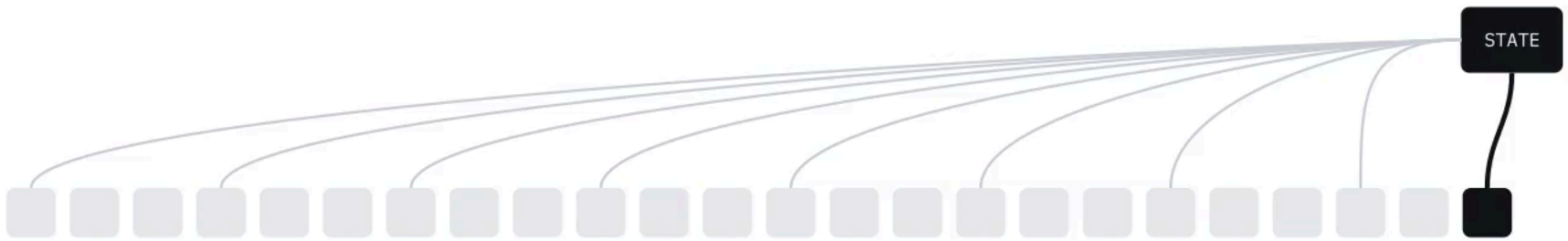
Source: model configs, read October 1st and 4th, 2026 · MiniMax M2 report, Table 2

 · © 2026 Julien Simon · CC BY 4.0

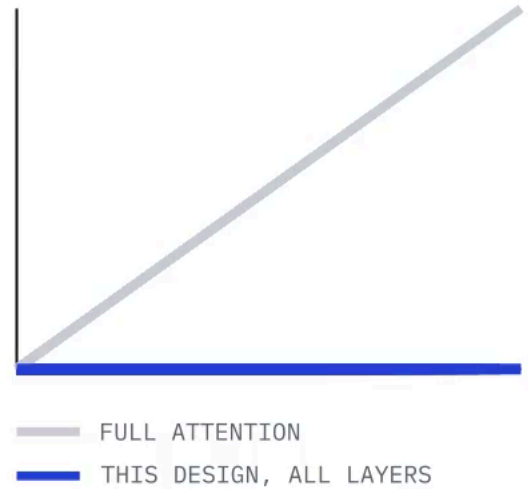
STEP 4 · MAY 2024

Mamba-2: no cache, one fixed-size state

STATE LAYERS · EACH TOKEN IS FOLDED INTO THE STATE



CACHE HELD AS TOKENS ARRIVE



SAVES	memory and compute: nothing grows with the context
GIVES UP	exact lookup: no earlier token is left to go back to
SHIPS IN	NVIDIA Nemotron 3 Ultra: 48 Mamba-2 layers to 12 attention

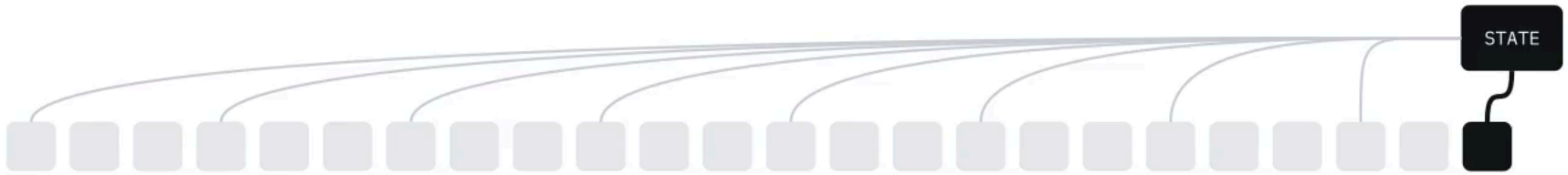
Source: Dao and Gu, 2024 · Nemotron 3 Ultra config

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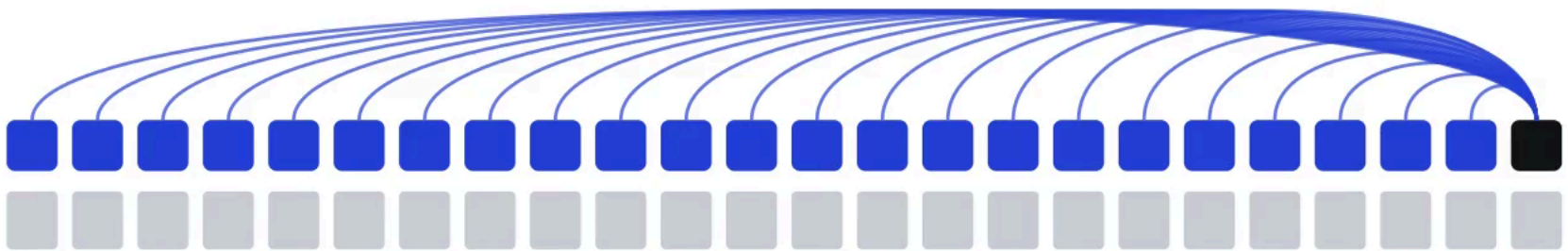
STEP 5 · SEPTEMBER 2025

Linear hybrid: three state layers, then one full layer

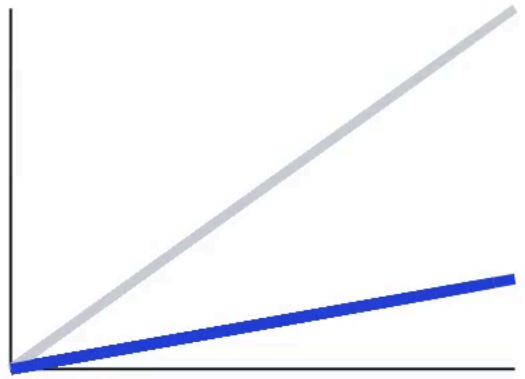
3 LAYERS OF 4 · LINEAR: A FIXED STATE



1 LAYER OF 4 · FULL ATTENTION



CACHE HELD AS TOKENS ARRIVE



LAYER MIX



Kimi K3 69 : 24
Qwen3.8 69 : 23
Qwen3.8-27B 48 : 16

SAVES	memory and compute in three layers of four
GIVES UP	exact lookup there: the pure layer scored 30.6 against 37.0
SHIPS IN	Qwen3-Next, Qwen3.5, Qwen3.8, Kimi K3

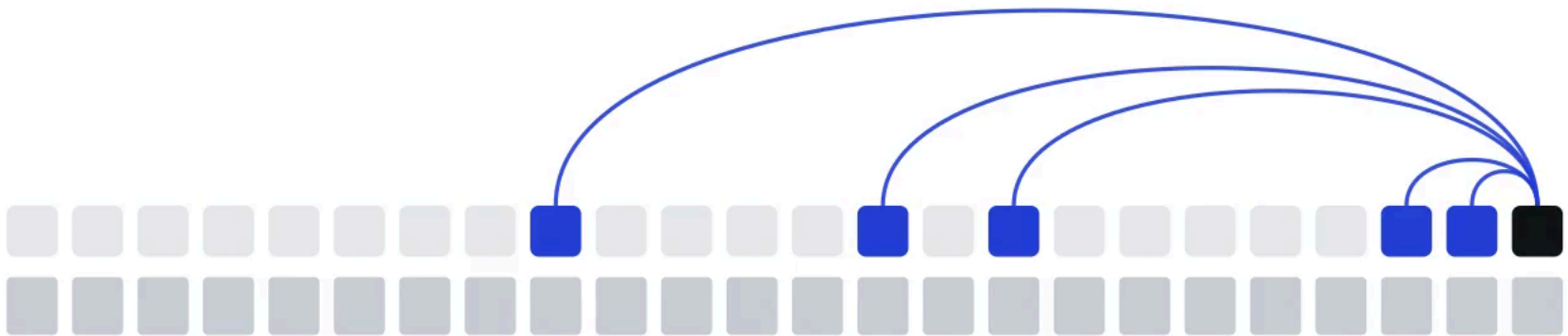
Source: Qwen3.8 and Kimi K3 configs · Gated DeltaNet paper, Table 4

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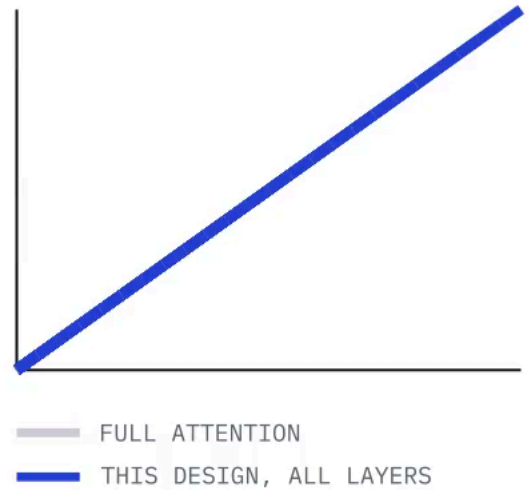
STEP 6 · SEPTEMBER 2025

Sparse attention: an indexer picks what to read

EVERY LAYER · THE INDEXER SCORES EVERY TOKEN, THE LAYER READS THE BEST 2,048



CACHE HELD AS TOKENS ARRIVE



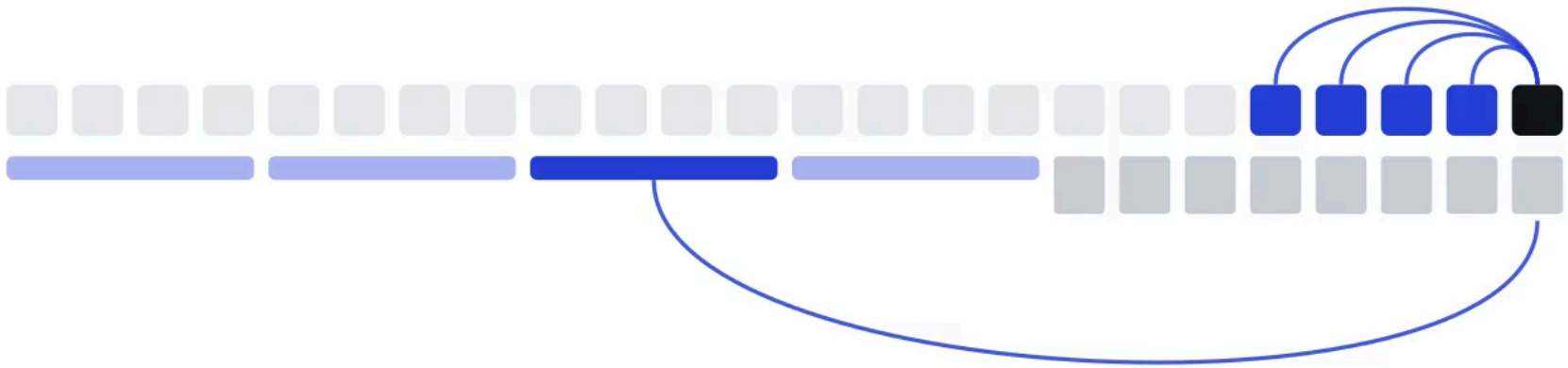
SAVES	compute, though the indexer still scores every token
GIVES UP	whatever the indexer misses; the cache is not smaller
SHIPS IN	DeepSeek V3.2, GLM-5.3 (all 78 layers), MiniMax M3

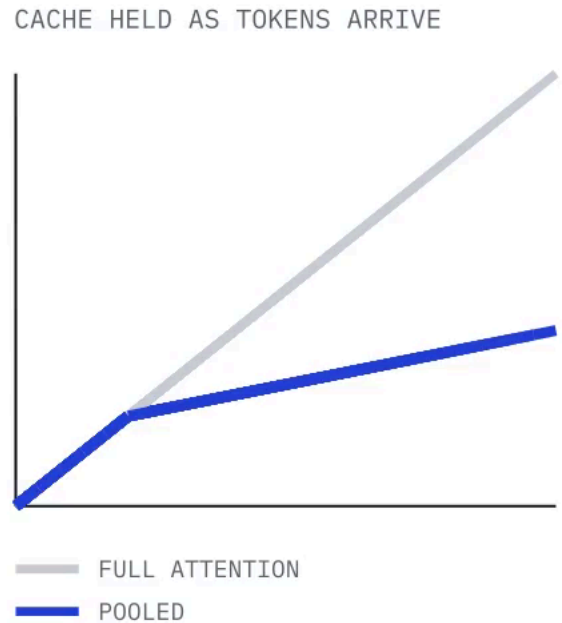
Source: DeepSeek V3.2 report · GLM-5.3 config

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STEP 7 · DEEPSEEK V4 · APRIL 2026

Pooled cache: old tokens become summaries





GREY CELLS	the cache kept for each token; merged four into one as they age
ARCS ABOVE	the most recent tokens, read in full (128 in V4)
ARCS BELOW	summaries of older tokens, a few picked by an indexer
DEEPSEEK'S ESTIMATE	V4-Pro at 1M tokens: 10% of V3.2's cache, 27% of its compute

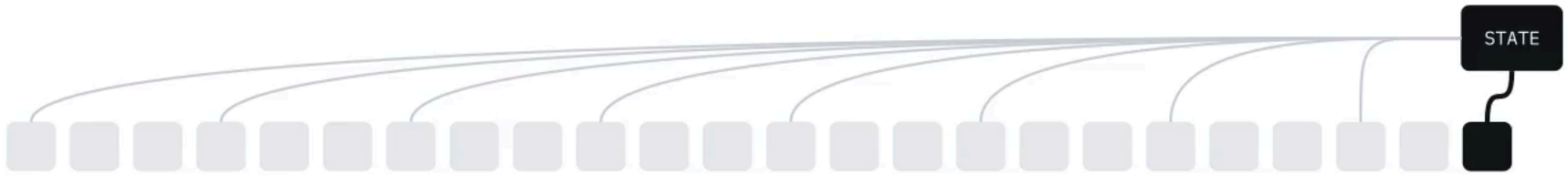
Source: DeepSeek-V4 report, April 2026 · 24 tokens stand in for a long context

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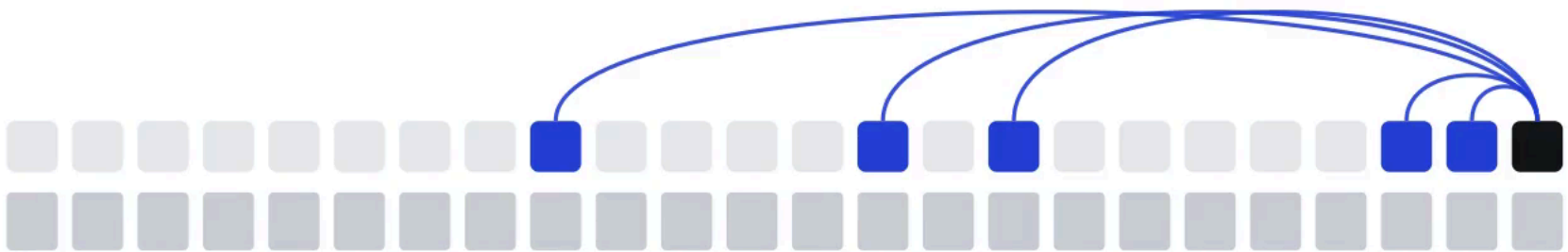
STEP 8 · AUGUST 2026

Linear plus sparse: no layer reads everything

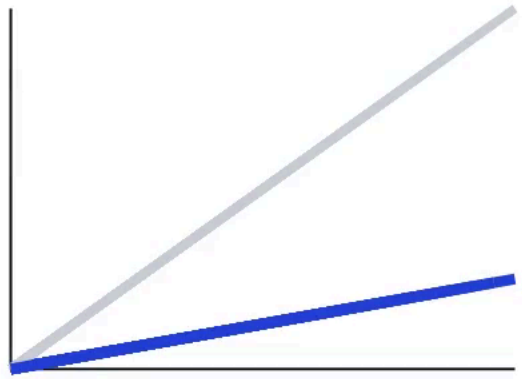
3 LAYERS OF 4 · LINEAR: A FIXED STATE



1 LAYER OF 4 · SPARSE: TOKENS PICKED BY AN INDEXER



CACHE HELD AS TOKENS ARRIVE



LAYER MIX



Qwen3.8-Flash-Next 36 : 12
GLM-5.3-Flash 34 : 11

SAVES	memory and compute, in every layer
GIVES UP	the last layer that read every token
SHIPS IN	Qwen3.8-Flash-Next, GLM-5.3-Flash

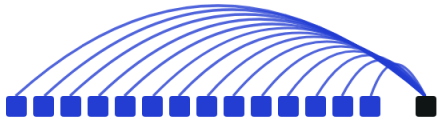
Source: Qwen3.8-Flash-Next and GLM-5.3-Flash configs, August 26th, 2026

 · © 2026 Julien Simon · CC BY 4.0

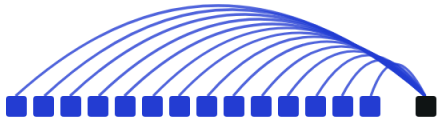
THE EIGHT STEPS

What the newest token reads, design by design

1 · GROUPED-QUERY
every token, smaller cache



2 · LATENT (MLA)
every token, compressed



3 · SLIDING WINDOW
the last 128 to 2,048



4 · MAMBA-2
a fixed state



5 · LINEAR HYBRID
state in 3 layers, all in the 4th



3 layers of 4 above · the 4th below

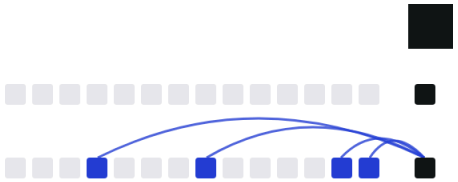
6 · SPARSE
2,048 picked



7 · POOLED
summaries + last 128



8 · LINEAR + SPARSE
state in 3, picks in the 4th

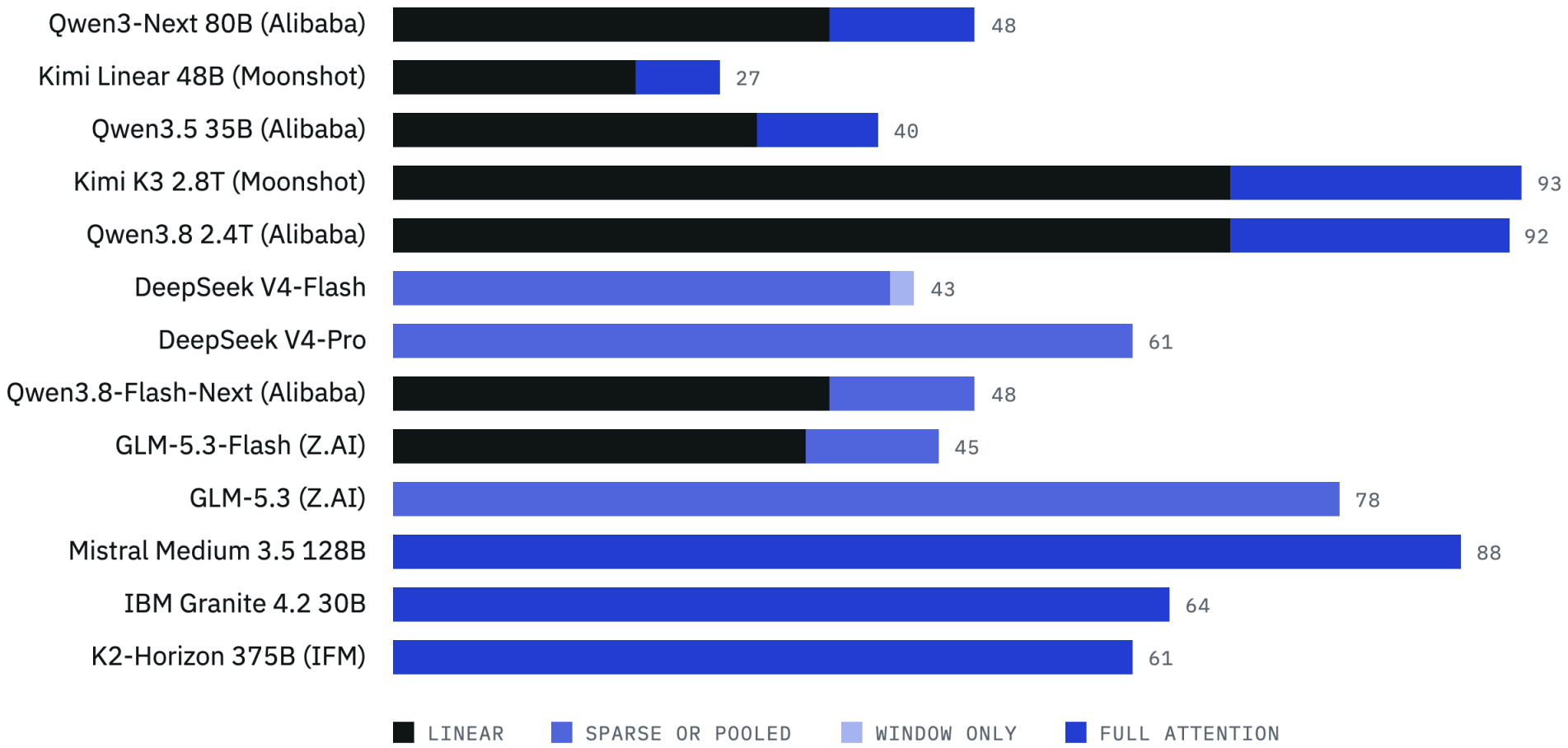


3 layers of 4 above · the 4th below

Source: the piece's eight steps; dates and models on the slides that follow

WHAT THE CONFIG FILES SAY

Layers by type, read from each model's config.json



Source: config.json of each model on Hugging Face, read October 1st and 4th, 2026

Who tested it against full attention?

Ten published comparisons, six labs, one rule applied to **every result each paper reports**:

- **win**: more than a point ahead somewhere, never more than a point behind
- **tie**: within a point everywhere
- **mixed**: more than a point apart in both directions

Result: **1 win, 7 mixed, 2 losses**

The papers behind each row are in note 20 of the piece

WHO TESTED IT AGAINST FULL ATTENTION

Ten comparisons, judged on every result each paper reports

WINS	**Z.AI, Feb 2026 · GLM-4.7-Flash, 31B**	Sparse, converted
MIXED	**Alibaba, Aug 2026 · 25B, 3B active**	Gated DeltaNet hybrid
MIXED	**Ai2, Mar 2026 · 7B**	Gated DeltaNet hybrid, 3 to 1
MIXED	**Moonshot, Oct 2025 · 48B, 3B active**	KDA hybrid, 3 to 1
MIXED	**MiniMax, Jun 2026 · 109B, 6B active**	Sparse, top 2,048, converted
MIXED	**DeepSeek, Sep 2025 · V3.2-Exp**	Sparse, top 2,048, converted
MIXED	**Z.AI, Feb 2026 · GLM-5 base model, 744B**	Sparse, from a dense parent
MIXED	**Alibaba, Aug 2026 · 125B, 6B active**	Sparse fourth layer
LOSES	**Z.AI, Feb 2026 · 9B**	Gated DeltaNet conversion, 1 to 1
LOSES	**MiniMax, May 2026 · M2 scale**	Sliding-window conversion

1 win · 7 mixed · 2 losses

win: more than a point ahead somewhere, never more than a point behind

Source: each lab's paper or report, cited in note 20 of the piece

What is missing

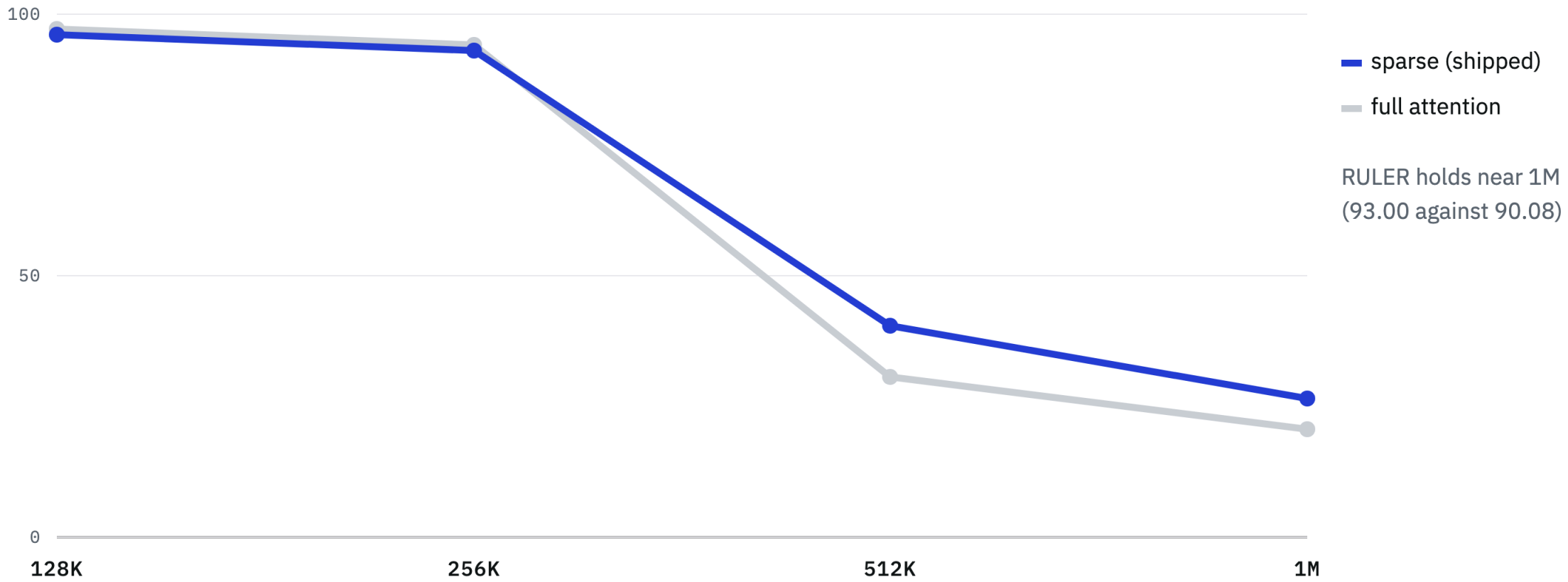
1. **Size**: no flagship compared with an all-full-attention model of its own size
2. **Task**: needles are where sparse did well; reranking, tracking and QA are closer to agent work
3. **Depth**: RULER holds near a million; telling **eight** similar things apart gives out between **256K** and **512K**, with either fourth layer

Alibaba's eight-needle curve: 96 · 93 · **41** · 26 (128K, 256K, 512K, 1M)

Qwen3.8-Flash-Next technical report, Table 3, August 26th, 2026

DEPTH

Telling eight similar things apart gives out past 256K



Source: Qwen3.8-Flash-Next technical report, Table 3, MRCR eight-needle, August 26th, 2026

Measured on this Mac: three August 2026 models, ~30B each

	Design	Layers
Granite 4.2 30B	full attention	64 full
Muse Glimmer 30B	sliding window	39 window (2,048) + 13 full
Qwen3.8-27B	linear hybrid	48 linear + 16 full

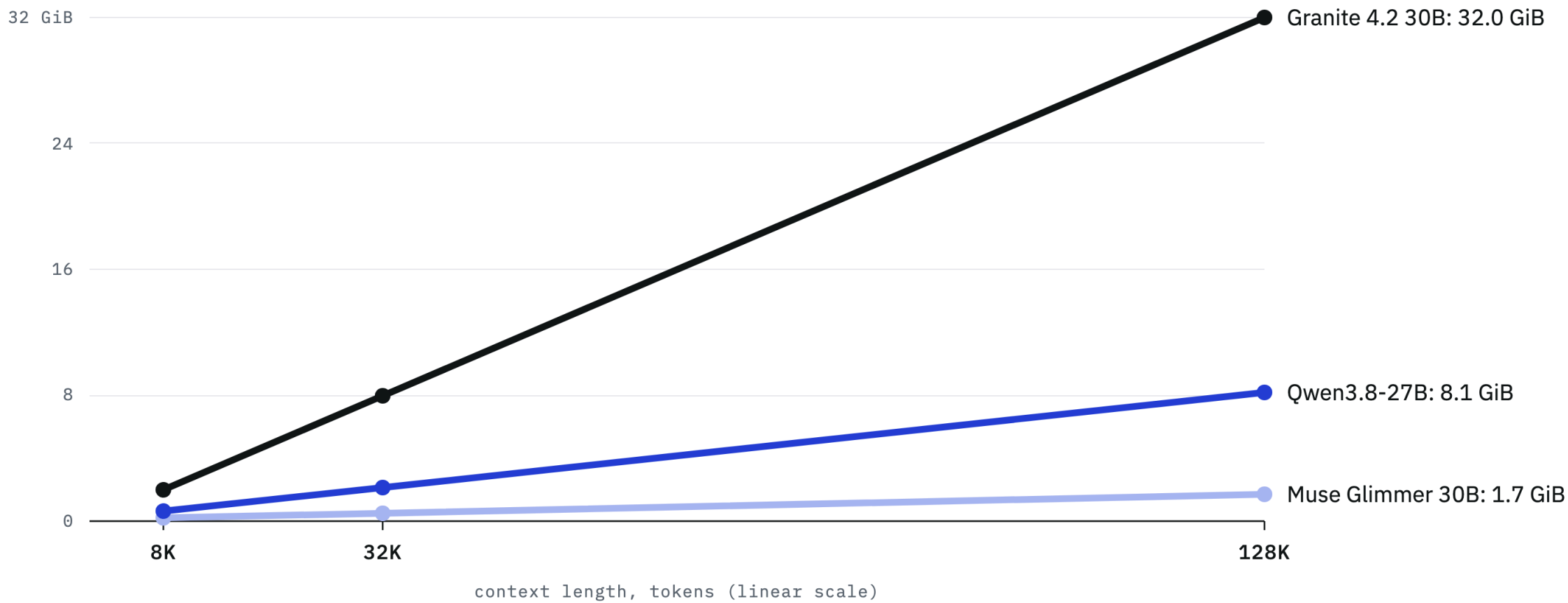
Cache memory at 8K, 32K and 128K · decode speed at depth

Configs on Hugging Face, read October 4th, 2026 · bench/README.md for the method

MEASURED ON THIS MAC

Cache memory at each context length

KV cache plus recurrent state, as llama.cpp allocates it at start-up



Source: bench/results.csv: llama.cpp allocation log on this Mac (method in bench/README.md)

Where the cost reaches you

- **Cache**: a hybrid's cached prefix needs a saved state *and* KV at the same position; ~49 MiB per state on Qwen3.5-4B
- **Determinism**: on vLLM, cached and fresh Gated DeltaNet answers are not bit-exact; compare scores, not outputs
- **Hardware**: FlashMLA, September 30th, 2026: "we removed support for the Hopper architecture"
- **Concurrency**: "capped to 9 by the mamba state cache"

SuffixReplay arXiv 2609.33477 · docs.lmcache.ai · github.com/deepseek-ai/FlashMLA · SGLang #38846

Six checks before you build on one

1. Open `config.json`: count the layer types, read the keys next to "full attention"
2. Retrieval at 32K and at your length, target at several positions
3. Measure cache hit rate on your engine and version; compare scores, not outputs
4. Size concurrency against state memory as well as cache memory
5. Ask which GPU generation has a fast path in this release
6. Price the workload two ways, at the hit rate you measured

Offered at a million tokens.

Compared with an all-full-attention model at 128,000, at most. The stretch in between is yours to measure.

"Eight Changes to Transformer Attention, and What Each Costs", airealist.ai, October 2026
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